

CS 660: Data Mining For Decision Making

Lecture 1-3 (Week 1)

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ScalingtheHeights

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A Little About Me! - In the office

Qualifications

- M.S. degrees in Software Engineering, Engineering and Public Policy, and Rational Simulation (cognitive modeling) from Carnegie Mellon University
- Ph.D. in Engineering and Public Policy from Carnegie Mellon University
- Post-doctoral fellowship from Carnegie Mellon University
- Since 2012 at Indian Institute of Technology, Mandi, India

Research interests

- Artificial intelligence and cognitive modeling, Human-Computer Interaction, Environmental decision making, Judgment and Decision Making

Professional Experience

- Served as a Software Engineer in Tata Consultancy Services (TCS) and in MothersonSumi Infotech and Designs Ltd.
- Editorial Board Member, Scientific Reports (Nature Portfolio), contributing to peer review and editorial decisions in the field of computational and cognitive sciences.

Teaching Assistants

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What about youfolks?

- Please introduce yourselves

Announcements

- Syllabus
- Your Grade:
 - 30% Final Exam
 - 15% Mid-sem Exam
 - 10% Class Participation
 - 20% Class Assignments
 - 25% Class Project

Course Logistics

- **Please don't copy or plagiarize!**
 - **Being an AI researcher, I know how to catch it**
 - **If found, consequences will be catastrophic!**
- **If you did copy, then please cite the sources as (author, date). E.g., (Dutt, 2012)**

An Example (Witten, Frank, & Hall, 2011)

Human in vitro fertilization involves collecting several eggs from a woman's ovaries, which, after fertilization with partner or donor sperm, produce several embryos. Some of these are selected and transferred to the woman's uterus. The challenge is to select the "best" embryos to use—the ones that are most likely to survive. Selection is based on around 60 recorded features of the embryos—characterizing their morphology, oocyte, and follicle, and the sperm sample. The number of features is large enough to make it difficult for an embryologist to assess them all simultaneously and correlate historical data with the crucial outcome of whether that embryo did or did not result in a live child. In a research project in England, machine learning has been investigated as a technique for making the selection, using historical records of embryos and their outcome as training data.

Data Mining: What is it? (Witten, Frank, & Hall, 2011)

- Data mining is defined as the process of discovering structural patterns in data.
- The process must be automatic or (more usually) semiautomatic.
- The patterns discovered must be meaningful in that they lead to some advantage, usually an economic one.
- The data is invariably present in substantial quantities.

Example

Table 1.1 Contact Lens Data

Age	Spectacle Prescription	Astigmatism	Tear Production Rate	Recommended Lenses
young	myope	no	reduced	none
young	myope	no	normal	soft
young	myope	yes	reduced	none
young	myope	yes	normal	hard
young	hypermetrope	no	reduced	none
young	hypermetrope	no	normal	soft
young	hypermetrope	yes	reduced	none
young	hypermetrope	yes	normal	hard
pre-presbyopic	myope	no	reduced	none
pre-presbyopic	myope	no	normal	soft
pre-presbyopic	myope	yes	reduced	none
pre-presbyopic	myope	yes	normal	hard
pre-presbyopic	hypermetrope	no	reduced	none
pre-presbyopic	hypermetrope	no	normal	soft
pre-presbyopic	hypermetrope	yes	reduced	none
pre-presbyopic	hypermetrope	yes	normal	none
presbyopic	myope	no	reduced	none
presbyopic	myope	no	normal	none
presbyopic	myope	yes	reduced	none
presbyopic	myope	yes	normal	hard
presbyopic	hypermetrope	no	reduced	none
presbyopic	hypermetrope	no	normal	soft
presbyopic	hypermetrope	yes	reduced	none
presbyopic	hypermetrope	yes	normal	none

Structural Description (Pattern) in Data

If tear production rate = reduced then
recommendation = none

Otherwise, if age = young and astigmatic =
no then recommendation = soft

Weather Dataset

Table 1.2 Weather Data

Outlook	Temperature	Humidity	Windy	Play
Sunny	hot	high	false	no
Sunny	hot	high	true	no
Overcast	hot	high	false	yes
Rainy	mild	high	false	yes
Rainy	cool	normal	false	yes
Rainy	cool	normal	true	no
Overcast	cool	normal	true	yes
Sunny	mild	high	false	no
Sunny	cool	normal	false	yes
Rainy	mild	normal	false	yes
Sunny	mild	normal	true	yes
Overcast	mild	high	true	yes
Overcast	hot	normal	false	yes
Rainy	mild	high	true	no

In this case there are four attributes: *outlook*, *temperature*, *humidity*, and *windy*. The outcome is whether to play or not.

Structural Description (Pattern) in Data (also, called a Decision List)

- A set of rules learned from this information might look like this:
- If outlook = sunny and humidity = high then play = no
- If outlook = rainy and windy = true then play = no
- If outlook = overcast then play = yes
- If humidity = normal then play = yes
- If none of the above then play = yes

Decision List

- These rules are meant to be interpreted in order:
- The first one; then, if it doesn't apply, the second; and so on. A set of rules that are intended to be interpreted in sequence is called a *decision list*.
- Interpreted as a decision list, the rules correctly classify all of the examples in the table, whereas taken individually, out of context, some of the rules are incorrect. For example, the rule *if humidity = normal then play = yes* gets one of the examples wrong (check which one).

Weather Dataset: Two of the attributes—temperature and humidity—have numeric values

Table 1.3 Weather Data with Some Numeric Attributes

Outlook	Temperature	Humidity	Windy	Play
Sunny	85	85	false	no
Sunny	80	90	true	no
Overcast	83	86	false	yes
Rainy	70	96	false	yes
Rainy	68	80	false	yes
Rainy	65	70	true	no
Overcast	64	65	true	yes
Sunny	72	95	false	no
Sunny	69	70	false	yes
Rainy	75	80	false	yes
Sunny	75	70	true	yes
Overcast	72	90	true	yes
Overcast	81	75	false	yes
Rainy	71	91	true	no

Structural Description (Classification Rules)

For this example, there must be inequalities involving these attributes rather than simple equality tests as in the former case.

This is called a *numeric-attribute problem*—in this case, a *mixed-attribute problem* because not all attributes are numeric.

Now the first rule given earlier might take the form

If outlook = sunny and humidity > 83 then play = no

Association Rules

The rules we have seen so far are *classification rules*: They predict the classification of the example in terms of whether to play or not. It is equally possible to disregard the classification and just look for any rules that strongly associate different attribute values. These are called *association rules*. Many association rules can be derived from the weather data in Table 1.2. Some good ones are

If temperature = cool	then humidity = normal
If humidity = normal and windy = false	then play = yes
If outlook = sunny and play = no	then humidity = high
If windy = false and play = no	then outlook = sunny and humidity = high

Association Rules

All these rules are 100% correct on the given data; they make no false predictions. The first two apply to four examples in the dataset, the third to three examples, and the fourth to two examples. And there are many other rules. In fact, nearly 60 association rules can be found that apply to two or more examples of the weather data and are completely correct on this data. And if you look for rules that are less than 100% correct, then you will find many more. There are so many because, unlike

classification rules, association rules can “predict” any of the attributes, not just a specified class, and can even predict more than one thing. For example, the fourth rule predicts both that *outlook* will be *sunny* and that *humidity* will be *high*.

Preparing Input Data for Data Mining

- Data Cleaning (scrubbing, also called data cleansing), is the process of amending or removing data in a database that is incorrect, incomplete, improperly formatted, or duplicated. It is a time consuming activity often done in a semi-automated manner.
- **Missing Values:** Missing values are frequently indicated by out-of-range entries. Example: A negative number (e.g., -1) in a numeric field that is normally only positive, or a 0 in a numeric field that can never normally be 0. For nominal attributes, missing values may be indicated by blanks or dashes.
- **Inaccurate Values:** Pepsi somewhere and Pepsi-Cola somewhere else. Typographical errors. Example: Super-market seller uses her own cards for discounts to those who forgot their cards.

Applications of Data Mining in Real World

- Web-mining: Prestige of a web-page based upon how many link to it (PageRank)
- Decisions involving judgments (Banks use data-mining while giving you loans – accept or reject cases)
- Screening images (oil slicks or not in sea using satellite data)
- Load forecasting in Electricity Industry
- Diagnosing faults in machines in Industry
- Marketing and Sales (Pharmaceutical Industry – Patient Journeys, Market-Basket Analysis (Pepsi and Diapers on Thursdays), Discount or Loyalty Cards to Collect Data

Activities

- Read Witten, Frank, and Hall, 2011: Chapter 1 (up to page 15 before CPU performance; 21-29, 51-52, 58-60):
https://www.dropbox.com/s/ip50rwdynqnmdwb/%5BIan_H._Witten%2C_Eibe_Frank%2C_Mark_A._Hall%5D_Data_Mining%28BookFi.org%29.pdf?dl=0
- Read Singhal, 2011

Thank you!

Comments and Questions most welcome!